

2) $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ find $\lambda = ?$

$\rightarrow |A - \lambda I| = 0$

$$\left| \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} (4-\lambda) & (1-0) \\ (2-0) & (3-\lambda) \end{vmatrix} = 0$$

$$(4-\lambda)(3-\lambda) - (2-0)(1-0) = 0$$

$$(12 - 4\lambda - 3\lambda + \lambda^2) - (2 \times 1) = 0$$

$$(12 - 7\lambda + \lambda^2) - 2 = 0$$

$$10 - 7\lambda + \lambda^2 = 0$$

$$\therefore \lambda^2 - 7\lambda + 10 = 0$$

$$\lambda^2 - 5\lambda - 2\lambda + 10 = 0$$

$$\lambda(\lambda - 5) - 2(\lambda - 5) = 0$$

$$(\lambda - 5) = 0 \quad \text{OR} \quad (\lambda - 2) = 0$$

$$\therefore \underline{\lambda = 5}$$

$$\underline{\lambda = 2}$$

* vector :-